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THE IMPACT OF BUSINESS INTELLIGENCE ON INCREASING THE EFFICIENCY AND HEALTH QUALITY OF SURGICAL CENTERS

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Abstract. There are many reasons why healthcare organizations are choosing business information systems in their organization. One of the reasons is the steady growth of the health industry, and the medical centers are constantly looking for new ways to increase productivity and reduce costs. The purpose of this study was to investigate the impact of trade on increasing the productivity and health quality of surgical centers. This study is applied and will be done by the descriptive cross-sectional method. The statistical population of the research is Velenjak Hospital. There are 146 members for estimating the sample size using the Cochran sample size formula; based on this sample size, 9 clinics are estimated for the study. A questionnaire with Cronbach's alpha coefficient of 0.92 was used for data collection. The data of the present research were extracted from a questionnaire and a central registration database. These data are then converted to the relevant scores and prepared using Microsoft Power BI software for the analysis and used for investigating research hypotheses. The results showed that with the help of business intelligence, the quality of health can be improved and also the impact of business intelligence on the development of the surgical center leads to high efficiency.
Keywords: Business Intelligence, Increasing Efficiency, Health Quality

INTRODUCTION

Large health centers, especially hospitals, are important places in the health care field where it is important to examine the process of information being produced and transmitted from one sector to another. But the massive production of information and their heterogeneous management is an important challenge in this field. One of the problems facing hospital information systems is the lack of accurate and up-to-date patient data when needed. Nowadays, hospital organizations need the right information at the right time (Parsi, Kiani Nikou & Mahmoudi, 2016; Ghasemian, Fazlollah Tabar, 2016).

However, many of the systems used do not perform well. With the advent of health business intelligence systems, health concerns can be largely eliminated. The term business intelligence was first coined by Lohen, a researcher at IBM. He described business intelligence as a system consisting of data processing machines for the automatic abstraction of documents and the creation of expected profiles for each of the actions taken in an organization (Božič, 2019). Then, in the late 1990s, the term was used by Dresner, Gartner group analyst. He defined business intelligence, as including concepts and methods for improving business decision making using the support of reality-based systems (Pareek, 2006, Cody W. 2002).

Some researchers believe that the term business intelligence can be used when it refers to an organization's information and knowledge domains and describes the organization's environment, market conditions, customers, competitors, and economic issues. Because the systematic processes with the organization they provide and with the analysis and dissemination of information make the right decisions (Turban, 2010).

In recent years, in the field of health care, due to the massive amount of information produced and the variety of information, a tool called the hospital information system has been used. The hospital information system plays an important role in providing health care services and enables the improvement and expansion of communication between staff and departments within the hospital and with the organizations outside the hospital (Kurnia, 2018). Health organizations need to be

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smart. Therefore, implementing business intelligence systems has become an essential aspect of hospitals, especially hospitals in developing countries (Sabherwal, 2011).

Acquiring, deploying, transferring and retrieving knowledge and information within the organization ensures the survival of the organization and makes organizations operate smartly. Business intelligence systems improve the business efficiency of organizations through the correct and advanced use of customer information, vendors and internal business operations of organizations (Poon, 2004).

What is certain is that medical centers and private hospitals need to compete more closely with other medical centers because of their organizational nature and how they achieve financial and managerial independence. This necessitates the use of new technology-based management approaches to gain insights into the performance of units and Organizational evaluation of these centers. This points to the importance of things such as enhancing data quality and having an integrated measurement system that are considered in the senior information technology management by NGOs, including health centers. In the private health care centers, due to the attraction of customers and creating loyalty in them, it is also important to provide a better and more desirable service in order to maintain a competitive advantage. Health centers need a lot of readiness to execute the business intelligence projects of hospital health. The purpose of this study was to investigate the impact of business intelligence on increasing the efficiency and health quality of surgical centers.

RESEARCH BACKGROUND

In 2017, Gardbo et al examined the evaluation of business intelligence in public hospitals. Their purpose is to investigate the factors that influence business success (BI). They found that the quality of the system was positively and significantly associated with user satisfaction and that the quality of information was positively and significantly associated with the user satisfaction. User satisfaction is positively and significantly related to individual influence. The findings also provide empirical support for the role of user satisfaction as a mechanism that mediates the relationship between information quality or system quality and individual impact. User satisfaction is not only a critical structure in the success model of information systems, but also acts as a mediator (Shen, 2017).

In 2017, Chi Shan et al studied the maturing of business intelligence in hospitals. Information managers agreed that fast and accurate decision making is essential for organizational agility and that information plays an important role in the decision-making process. In 1997, Naimi Rad et al explained the necessity of using business intelligence in the business system. Business intelligence is a term used in business management to describe applications and technologies that are used to integrate, access, and analyze data and organizational information in order to help make better business decisions. In the past, business intelligence systems were designed with a focus on the back-end, but today the architecture of business intelligence has shifted to enterprise organizational analytic and in multilayer distributed applications. Business intelligence means giving the right information to the right people at the right time to make the right decision. In 2015, Ghalandri et al studied the dimensions of business intelligence and performance of health care executives. The results show that there is a significant relationship between the business intelligence (with the dimensions of strategic insight, shared destiny, desire for change, morale, unity and agreement, organizational pressure) and the performance of health center executives; in terms of the application of knowledge and organizational intelligence, no significant relation was observed. Abdi et al studied the application of business intelligence in healthcare in 2010. In this research, the role and position of business intelligence in the health sector have been investigated. After examining external and internal studies, a conceptual model was designed for a teaching hospital. In this study, observation and checklists were used to collect information and were used by computer and management experts in the data warehouse design. In a 2014 research, Habibi et al examined the relationship between business intelligence and customer relationship management and stated that the managers need intelligence, not information, for the decision-making process. Descriptive and inferential statistics

were used for data analysis. Also, a single sample was used test results of which show that business intelligence has a significant relationship with customer relationship management. In 2016, Qasemian et al investigated the business intelligence model for managing health systems using the user communication network. In this model, they attempted to develop a model for evaluating and managing the health system using proposals and tools such as the User Communication Network and Multilevel and Multi-opined Framework (MTMl).

THEORETICAL FOUNDATIONS OF RESEARCH

Business Intelligence: The concept of business intelligence is relatively new, but is associated with a wide range of techniques, tools and applications (such as online transaction processing, online analytical processing, analytical database, data mining, intelligent decision making system, knowledge management system, supply chain management, Customer Relationship Management and Enterprise Resource Planning); it is aimed at improving the quality of operations and the analysis of operations. In addition, business intelligence is the process of transforming data into information, in a way that enables the business organization to analyze information so that it can meet the needs of all beneficiaries of the system with the proper insight and understanding (Delawari et al., 2015).

Business Intelligence (BI) systems have become an integral part of the organization in the 21st century due to the ever-increasing need of business organizations for data analysis, data interpretation and processing. The definition of business intelligence refers to a broad concept of business intelligence that is designed to support and improve decision-making, which in turn leads to increased productivity in an organization (Remigiusz Tunowski, 2015).

Butler (2001) argues that business intelligence is the correct and effective management of information for a business purpose and is primarily concerned with the relationship between the user and the IT systems. Business intelligence has been introduced not as a tool and a product or even a system, but as a new approach in the organization architecture based on the speed of information analysis in order to make accurate decisions in the least possible time (Shakeri et al., 2013). The most important goal of business intelligence is to enable cross-functional and interactive (often timely and immediately) access to data as well as to utilize this data and ultimately to enable managers and business analysts to conduct analytics appropriately (Habibi, 2015).

Business intelligence refers to the process of converting raw data into business and management information, which helps the decision-makers of the organization to make their decisions better and faster and to act on the right information (Lahrmann et al., 2011). The data are processed and transformed into knowledge by logging into the business intelligence system, then the obtained knowledge is analyzed and its analytical results are used by managers in their decision making and measures are taken to improve the performance of the organization.

The effectiveness of the business intelligence system shows that it achieves its intended goals and meets the information needs of the organization. The effectiveness of business intelligence is resulted from comparing the results with the predetermined goals of the business intelligence in the organization (Cheng, 2009).

The stages of implementing the business intelligence system in an organization include:

• Regulation and planning: This is a well-planned and timed program for gathering information in the organization.

• Data Collection: There are different sources for each organization. In this section, different sources of information should be explored in order to obtain useful information appropriate to the organization's questions and needs.

• Data Processing: This part of the raw data is converted in a way that can be analyzed. This could be done by creating a database, adding data to the current database or using graphs and diagrams. This is also called extraction, modification or loading.

• Analysis and Production: At this stage, the information extraction software and tools (data mining, analytical processing, and online transactions) are used to

classify and analyze data, and the results are presented as accurate and multifunctional reports.
• Publication: At this stage, the results are presented to the applicant in the form of analytical and multidimensional reports. Results are often reflected in the information dashboard.
Main components of the business intelligence system
In a business intelligence solution provided for an organization, different people are involved in different sectors: they must use different software and technologies at the different stages including gathering, storing and analyzing information and presenting the results obtained. The main components of a business intelligence system include the following cases (Bakhshandeh & Rahmati, 2016):
• Extraction, Transfer and Load (ETL)
• Data warehouse (DW)
• Systems of On-line Analytical Processing (OLAP)
• Data Mining (DM)
• Software of taking Report
• Smart Decision-helping System
• Smart Agent
• Knowledge Management System
• Customer relationship system
Features of Business Intelligence System
A business intelligence system based on the standard has certain features: The standard includes functionality, usability, reliability, efficiency and support capabilities. Based on what has been announced by the Gartner Institute, the following can be generalized and applied to the business intelligence systems (Mirabi et al., 2010):
• Functionality: A business intelligence system based on this feature should be able to answer: budgeting, planning, monitoring, forecasting, integration, financial reporting, credit card, dashboard, in-depth reports, in-depth reports up to seed Information, analysis, user-defined querying, time and periodic taking report, and what-if analyses.
• Usability: To support this feature, we should have the following features: Ability for the user to access data at the level of details and summarized levels. The possibility for the user to create their own graphical environment. Ability to submit reports in various forms including graph, table, map, document and so on.
• Reliability: A business intelligence system should be able to provide reliable access regardless of hardware or number of users. Having lost some of the data, the system must be able to make accurate estimates of the missing values so that business intelligence software can continue to function properly.
Efficiency: In order for a business intelligence system to perform its intended operation correctly and efficiently, it must be able to use a number of tools including: (Mulit Relational Dimensional Data) MRDB, MDDBMulit Dimensional Data Base, Data Warehouse, Double layer architecture, three layers and hyper data.
• Support Capability: The system provided by a vendor should be fully supported. Good support of a system and presentation of new versions by the provider as well as providing useful training materials related to software is one of the features of a business intelligence system.

RESEARCH METHODOLOGY

This research is applied in terms of purpose and it is a descriptive survey that will be done by a cross-sectional method. The statistical population of the research is Velenjak Hospital. The number of the population is 146 people. For estimating the sample size, the formula of "Cochran Sample Size" is used; so 9 clinics are estimated as sample size for our study. Data collection is done by library and field methods. In the field, a researcher-made questionnaire will be used to collect the research data. The questions are closed-ended and the five-option Likert scale is used to measure the criteria. To assess the content validity, the questionnaires were provided to the relevant experts including hospital top managers, physicians and stakeholders to provide their comments and suggestions on the content of the questionnaire items appropriate to the purpose of the research. The reliability of the

questionnaire was calculated by Cronbach's alpha, which was 0.92 for the whole questionnaire. The Hospital Satisfaction Questionnaire consisted of 11 items that were used to measure patients' satisfaction in the hospital. Questionnaire scoring is a 4-point Likert scale that scores 1, 2, 3 and 4 for the “completely unsatisfied", "unsatisfied", "satisfied" and “completely satisfied", respectively. To analyze your questionnaire score, summarize the scores obtained and then judge based on the basis of the table below. Note that the following points are for a questionnaire. For example, if you have 10 questionnaires, you must multiply the following points by 10.

Example: The lower limit of the questionnaire scores has been obtained as the following:

Number of Questionnaire’s questions * 1 = Low limit of Score

Low limit of Score	Moderate limit of Scores	High limit of Scores
11	28	44

Sum up your scores from the 16 expressions above. The minimum possible score will be 16 and the maximum 64.

Score between 11 and 17: Patients' satisfaction with the hospital is low.

Score between 17 and 28: Patients' satisfaction is moderate.

Score above 28: Patients' satisfaction with the hospital is high.

The data used by the software system applied at the center is based on the SQL SERVER database.

The information used includes patient admission information, discharge, insurance information, and clinic and pharmacy information. The main focus of the research is the admission information of 8022 people. The data in this research are extracted using the questionnaire and the Index Center database. This data is then converted to the related scores and prepared for analysis by Microsoft Power BI software and used to investigate the research hypotheses.

FINDINGS

Clinic	Number of admissions	Admission percentage per clinic
Gynecologist	2980	60.13%
General surgeon	1353	27.44%
Urology	324	6.57%
Anesthesiologist	134	2.72%
Pediatrician	116	2.35%
Internist	13	0.26%
Cardiovascular	7	0.14%
Orthopedics	3	0.06%
Practitioner	1	0.02%
Total	4931	100.00%

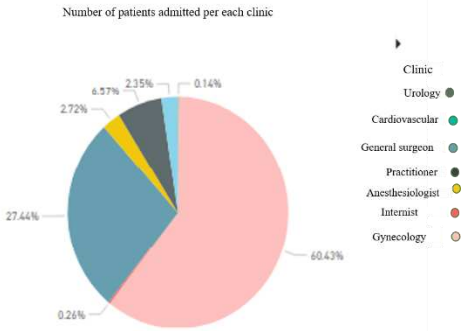


Chart 1. The number of patient admission according to the clinic

Chart1 shows that the obstetric and gynecologic clinic and general surgery had the highest acceptance with about 60% and 27% and the orthopedic and general clinics (less than 1% of total) had the lowest acceptance.

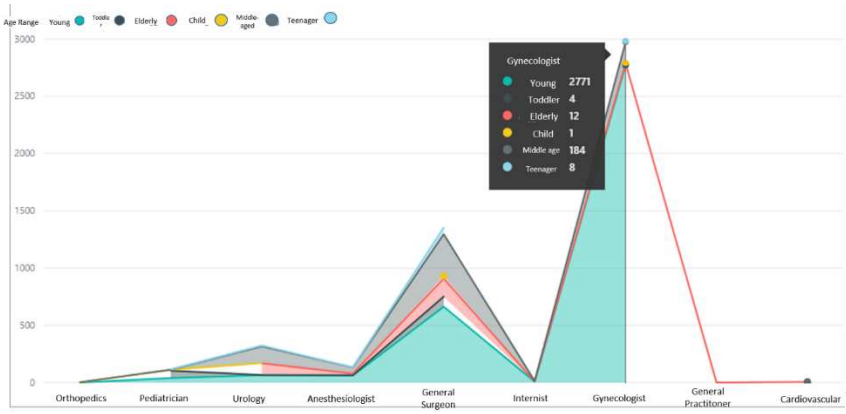


Chart 2.Number of admissions according to Clinic and Age range (area chart)

Received Amount for Each type of Insurance

Insurance Type	Received	Received of each insurance to total percentage
Keshavarzi Bank 2	202,000.00	0.01 %
Melli Bank	341,000.00	0.02 %
Dana	747,000.00	0.05 %
Sepah Bank	780,000.00	0.05 %
Mihan Insurance- electricity of west district	1,057,000.00	0.06 %
Refah Kargaran Bank	1,212,000.00	0.07 %
Retired Melli Bank	1,663,200.00	0.10 %
Sina	1,794,000.00	0.11 %
Mellat Bank	2,580,000.00	0.16 %
Mihan Insurance	2,608,000.00	0.16 %
Iran- Complementary insurance	3,190,000.00	0.19 %
Saderat Bank	3,439,200.00	0.21 %
Oil Industry	4,352,000.00	0.26 %
Iran	6,276,000.00	0.38 %
Health Services- others	6,918,900.00	0.42 %
Melli Bank-employees	7,005,600.00	0.42 %
Keshavarzi Bank	7,134,000.00	0.43 %
Tejarat Bank	9,244,400.00	0.56 %
Red Crescent Insurance	23,043,000.00	1.40 %
Health Services- Rural	50,925,280.00	3.09 %
Health Services- Iranian	64,880,370.00	3.93 %
Heath Services- Employees	121,858,820.00	7.38 %
Social Security	504,022,261.00	30.53 %
Free	825,405,500.00	50.00 %

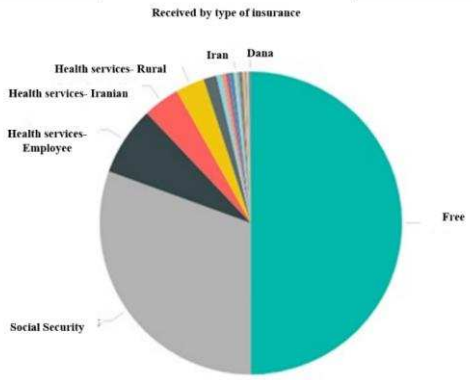


Chart 3. Amount received according to type of insurance

Chart3 shows that 50% of the services provided are covered by insurance.

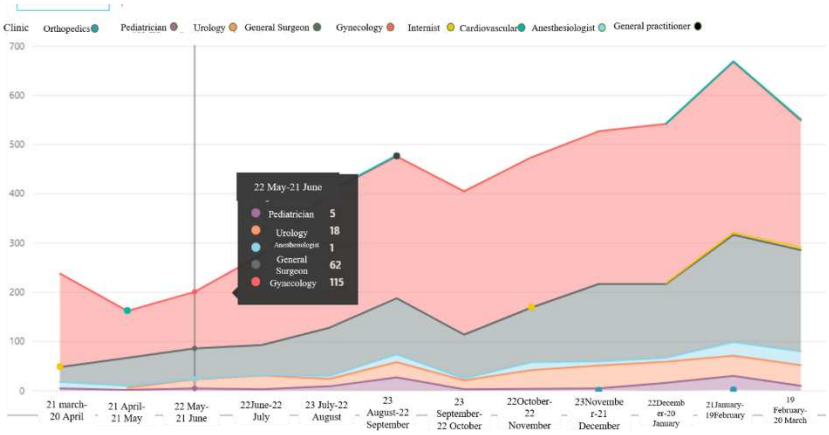
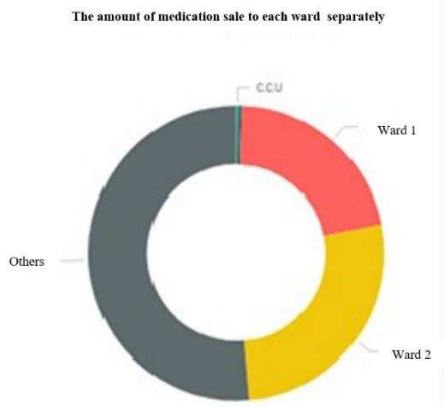


Chart 4. number of Admission according to Clinic and Month (Stacked Area Chart)

Chart 4 shows the number of admissions to the different clinics according to the month. The number of admissions in the second half of each year is clearly higher than the first half.



Ward	Drug Sales Price	Ward sales percentage to total
Others	258,495,539.50	51.48 %
Ward 2	133,911,411.00	26.67 %
Ward 1	106,608,555.00	21.23 %
C.C.U	1,633,800.00	0.33 %
Emergency	1,527,491.00	0.30 %
Total	**502,176,796.50**	**100.00 %**

Chart 5. Drug sales per hospital ward

The received cost to the number of admitted patients ratio

Clinic	Efficiency	Number of admissions	Received cost
Gynecologist	180,572.23		538,105,240.00
General surgeon	257,157.45	2253	347,934,041.00
Urology	1,787,280.99	324	579,079,040.00
Anesthesiologist	194,044.03	134	26,001,900.00
Pediatrician	182,014.74	116	21,113,710.00
Internist	128,623.08	13	1,672,100.00
Cardiovascular	1,187,900.00	7	8,315,300.00
Orthopedics	243,533.33	3	730,600.00
Practitioner	220,300.00	1	220,300.00
Total	308,897.23	4931	1,523,172,231.00

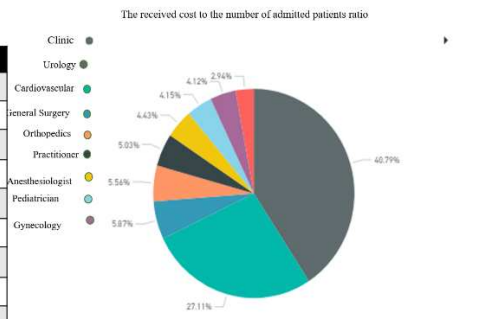


Chart 6. Ratio of the amount received according to admission per clinic

Chart 6 shows that urological and cardiovascular clinics (40 and 27 percent, respectively) have a good benefit in terms of the number of patients admitted.

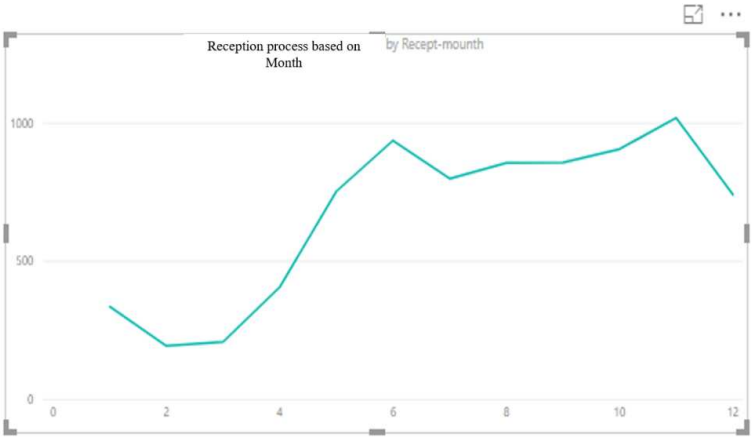


Chart 7. Acceptance trend and amount received per month

Chart 8. Monthly received amount

Chart 8 shows the increase in admission rate in the second half of the year, especially at three months of the end of the year.

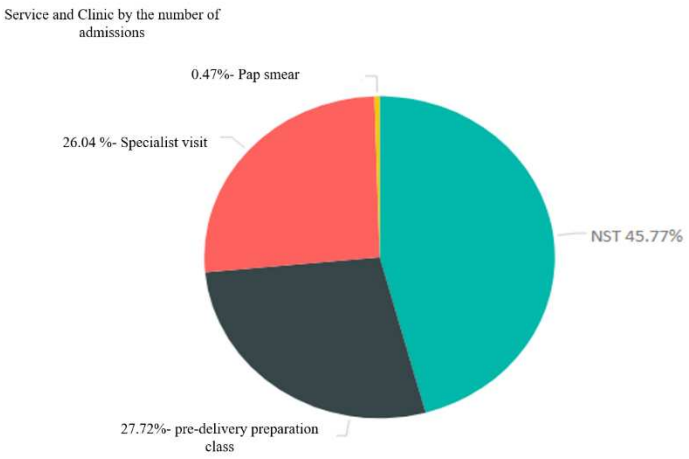


Chart 9. Number of services provided at the gynecologic clinic

Chart 9 shows that Pap smear test with 0.47% of admission, has the least referral among the services related to gynecology clinic. Pap smear test is the best way to diagnose cervical cancer. It is the fourth leading cause of cancer-related deaths among women worldwide and should be treated periodically.

CONCLUSION

The first hypothesis: Business intelligence can improve the quality of health. Considering the data for referrals at the Gynecology Clinic, we can see that the referral rate for Pap smear testing is very low (0.47%). For improving health, giving relevant brochures, we can inform the women about the necessity and sensitivity of this test.

The second hypothesis: Given the number of referrals to the different clinics and specialties, we can see the highest percentage of referrals to the gynecology clinic. Also, according to the chart of the ratio of income to the number of referrals, we can get a good return on the urology and cardiovascular clinics. Therefore, the development of the above clinics can generate considerable profits for the collection and is a profitable and low-risk investment.

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